Android Development with Kotlin

16-Week Study Plan & Syllabus

Prepared for **Anjana** | 4 hours daily | Beginner to App Publisher

Mentor's Rules

- **Write code every day.** Watching videos alone won't make you a developer. Type every example yourself.
- **Build, break and fix.** Errors are normal. Read the error message, Google it, and try again.
- **Use GitHub from Week 1.** Push your practice code there. It becomes your portfolio.
- **Learn one thing at a time.** Don't jump to advanced topics before you're ready.
- **Take 1 day off each week.** Rest helps you remember what you learned.

Daily Timetable (4 hours)

Time	Activity	What to do
Hour 1	Learn	Watch or read the day's new topic
Hour 2	Practice	Type the examples, change them, try small exercises
Hour 3	Build	Work on the weekly mini-project
Hour 4 (first 30 min)	Revise	Go over yesterday's topic and write short notes
Hour 4 (last 30 min)	Share	Push your code to GitHub and plan tomorrow

Take a 10-minute break after each hour. **Week plan:** study Monday to Saturday. Saturday = revise the week and finish the project. Sunday = rest, or light revision.

Syllabus (16 Weeks)

Phase 1: Kotlin Basics (Weeks 1-3)

Don't open Android Studio yet. Learn the language first.

Week	Topics	Mini-project
1	Install IntelliJ or use play.kotlinlang.org; variables (val/var); data types; operators; if/when; loops; functions	Simple calculator in the console
2	Null safety (?, ?., ?:, !!); collections (List, Set, Map); lambdas; higher-order functions; filter, map, forEach	Student marks manager (add, list, average, topper)
3	OOP: classes, objects, constructors, inheritance, interfaces; data class; enum; sealed class; extension functions; Git & GitHub basics	Library book system using OOP

Checkpoint: Can you write a small program with classes, lists and lambdas without looking at notes?

Phase 2: Android Basics with Jetpack Compose (Weeks 4-7)

Week	Topics	Mini-project
4	Install Android Studio; project structure; emulator; run on your phone; @Composable; Text, Image, Button; Modifier	Business card app
5	Layouts: Column, Row, Box; Material 3; themes, colors, fonts; LazyColumn (lists); cards	Contact list app
6	**State**: remember, mutableStateOf, state hoisting; TextField; user input; Activity lifecycle	Tip calculator, then a Dice Roller
7	Navigation Compose (multiple screens); passing data between screens; Scaffold, TopAppBar, BottomNav	Multi-screen quiz app

Checkpoint: Can you build a 2-3 screen app that reacts to button clicks and user input?

Phase 3: App Architecture & Data (Weeks 8-12)

Week	Topics	Mini-project
8	MVVM architecture; ViewModel; UI state; StateFlow; why architecture matters	Rebuild the quiz app with a ViewModel
9	**Coroutines**: suspend, launch, viewModelScope, Dispatchers; basics of **Flow**	Countdown timer app
10	**Internet**: REST APIs, JSON; Retrofit; Kotlinx Serialization; loading/error states; Coil (load images)	Weather app or Movie list app using a free API
11	**Local database**: Room (Entity, DAO, Database); DataStore (save settings); Repository pattern	Notes / To-Do app with saved data
12	**Hilt** (Dependency Injection); clean project structure (data, domain, ui packages)	Refactor the Notes app with Hilt

Checkpoint: Can you build an app that downloads data from the internet and saves it offline?

Phase 4: Professional Skills & Final Project (Weeks 13-16)

Week	Topics	Work
13	Permissions; notifications; WorkManager (background work); basic animations	Add reminders to your To-Do app
14	Testing: unit tests (JUnit), testing ViewModels, intro to Compose UI tests; debugging with Logcat	Write tests for your Notes app
15-16	**Final project**: plan, design, build, test	Expense Tracker, Recipe App or News App using Compose + MVVM + Room + Retrofit + Hilt
End of 16	Generate a signed APK/AAB; Play Store basics; write a good GitHub README with screenshots	Publish to GitHub, or to the Play Store if you can

After 4 Months: What Next?

- Build 2-3 more apps of your own, using your own ideas.
- Learn **Firebase** (login, cloud database).
- Explore **Kotlin Multiplatform** (Android + iOS from one codebase).
- Start applying for **internships**. Your GitHub projects are your resume.

Where to Study (Free Resources)

Main resource: Android Basics with Compose (by Google, free)
https://developer.android.com/courses/android-basics-compose/course
Starts from zero, teaches Kotlin first, then Android. Videos, codelabs, quizzes and projects. **If you use only one resource, use this one.**

Phase 1: Kotlin (Weeks 1-3)

Resource	Link / How to find	Use it for
Kotlin Playground	https://play.kotlinlang.org	Writing code in your browser
Official Kotlin docs	https://kotlinlang.org/docs/getting-started.html	Reading about each topic
Kotlin Koans	https://play.kotlinlang.org/koans	Small practice exercises
freeCodeCamp Kotlin course	Search YouTube: "freeCodeCamp Kotlin full course"	Complete course in one video

Phase 2: Android with Compose (Weeks 4-7)

Resource	Link
Android Basics with Compose (Units 1-4)	https://developer.android.com/courses/android-basics-compose/course
Jetpack Compose course	https://developer.android.com/courses/jetpack-compose/course
Android Studio download	https://developer.android.com/studio

Phase 3: Architecture & Data (Weeks 8-12)

Topic	Where to study
ViewModel, Coroutines, Retrofit, Room	Android Basics with Compose, Units 4-6
MVVM, Hilt, clean structure	Philipp Lackner on YouTube (search "Philipp Lackner MVVM" / "Philipp Lackner Hilt")
Official architecture guide	https://developer.android.com/topic/architecture

Phase 4: Professional Skills (Weeks 13-16)

Topic	Where to study
WorkManager, notifications	Android Basics with Compose, Unit 7
Testing	https://developer.android.com/training/testing
Real-world example app	https://github.com/android/nowinandroid
Publishing an app	https://developer.android.com/studio/publish

Other Helpful Resources

- **YouTube:** Philipp Lackner, Android Developers (Google official), freeCodeCamp.org
- **Git & GitHub:** https://learngitbranching.js.org and https://skills.github.com
- **Books (optional):** *Head First Kotlin* (beginner-friendly), *Kotlin in Action* (deeper, for later)
- **When stuck:** ask your mentor (Claude), https://stackoverflow.com, Reddit r/androiddev and r/Kotlin

Learning in Malayalam

Most good Android content is in English, and no reliable Malayalam course for Kotlin Android has been verified yet. Use these options:

1. Search YouTube with these words

- Kotlin Malayalam
- Android Kotlin Malayalam tutorial
- Jetpack Compose Malayalam
- Android Studio Malayalam

Choose	Avoid
Uploaded in 2023 or later	Very old videos (before 2021)
Uses Jetpack Compose	Uses Java or only XML layouts (old method)
A full series or playlist	Random single videos
Good comments, and the teacher replies	No comments, or comments turned off

2. Spoken Tutorial (IIT Bombay, free)

Android app using Kotlin - an Indian government-supported site with courses in many Indian languages. Check if Malayalam is available for this course. Good for the very basics.

3. English videos with Malayalam subtitles

On YouTube: **Settings** → **Subtitles/CC** → **Auto-translate** → **Malayalam**. Works well with Philipp Lackner and Android Developers videos.

4. Ask your mentor in Malayalam

Ask something like *"Explain null safety in Malayalam"* and you'll get a simple explanation with examples. Code stays in English.

Mentor tip: Use Malayalam videos to **understand the idea**, then use the Google course to **practice**. Slowly try more English content. It will help in interviews and when reading documentation.

Your First Task Today

Open https://play.kotlinlang.org and write a program that prints:

```
println("Hello, I am Anjana and I will become an Android Developer!")
```

Then learn the difference between **val** and **var**. Good luck, Anjana!